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Reg. No. | | | | | | | | | | | [14ME35]

THIRD SEMESTER B.TECH EXAMINATION DECEMBER - 2015

FLUID MECHANICS

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions. Taking at least select TWO questions from each part.
2. Assume any missing data suitably.

PART-A

1. a. Derive compressibility of a fluid. Prove that the compressibility for a perfect gas undergoing isothermal compression is 1/p while for a perfect gas undergoing isentropic compression is 1/γP where γ is the ration of two specific heats and P is the pressure. **06**

b. Two large horizontal fixed parallel planes are 12mm apart. The space between the surfaces is filled with oil of viscosity 0.972N-s/m^2. A flat thin plate of 0.25m^2 area moves through the oil at a velocity of 0.3m/s. Calculate the drag force when: **08**

- **i.** The thin plate is at the middle of the plane surfaces.
- **ii.** The thin plate is at a distance of 4mm from one of the plane surfaces.

2. a. State and prove Pascal's Law. **05**

b. Define: **05**

- **i.** Atmospheric Pressure.
- **ii.** Gauge Pressure.
- **iii.** Absolute Pressure.
- **iv.** Vacuum Pressure.

Represent these pressures on a chart.

c. The right limb of a simple U-tube manometer containing mercury is open to the atmosphere while the left limb is connected to a pipe in which a fluid of specific suavity 0.9 is flowing. The centre of the pipe is 12cm below the level of mercury in the right limb. Find the pressure of the fluid in the pipe if the difference of mercury level in the two limbs is 20cm. **05**

3. a. Define the following terms: **04**

- **i.** Metacentre.
- **ii.** Metacentric height.

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b. Explain the conditions of equilibrium of a floating body and submerged body. **05**

c. Determine the total pressure and centre of pressure of an isoscles triangular plate of base 4m and altitude 4m when it is immersed vertically in an oil of specific gravity 0.9. The base of the plate coincides with the free surface of oil.

4. a. Differentiate between: **03**

i. Steady and Unsteady flow.

ii. Laminar and turbulent flow.

iii. Compressible and incompressible flow.

b. Derive the three dimensional continuity equations for a steady incompressible flow in Cartesian co-ordinates. **05**

c. In a 2- dimensional incompressible flow, the fluid velocity components are given by $u = x - 4y$ and $v = -y - 4x$. Determine the velocity potential function and stream function. **06**

PART-B

5. a. Derive Euler's equation of motion along a stream line. Obtain Bernoulli's equation from Euler's equation. State all the assumptions. **07**

b. A 45° bend rectangular air duct of $1m^2$ cross sectional area is gradually reduced to $0.5m^2$ area. Find the magnitude and direction of the force required to hold the duct in position if the velocity of flow at $1m^2$ section us 12m/s and the pressure is $4N/cm^2$. Take the specific weight of air as $11.38N/m^3$. **07**

6. a. Explain how the discharge is measured using a pitot tube. **03**

b. Derive an expression for discharge over a triangular notes. **05**

c. A 30cm×15cm venturimeter is provided in a vertical pipe line carrying oil of specific gravity 0.8, the flow being upwards. The difference in elevation of throat section and entrance section of the venturimeter is 25cm. The differential U-tube mercury manometer shows a deflection of 30cm. Calculate: **06**

i. The discharge of oil.

ii. The pressure difference between the entrance and throat section.

Take c_d=0.98.

7. a. Prove that the velocity distribution across the cross section of a circular pipe is parabolic in nature. Also show that the maximum velocity is in the centre of the pipe and is equal to twice the average velocity. **08**

b. Water at 15°c flows between two parallel plates at a distance of 1.6mm apart. Determine: **06**

- **i.** Maximum Velocity.
- **ii.** Pressure loss per unit length.
- **iii.** Shear stress at the plate.

Take the average velocity of water as 0.2m/s. The viscosity of water at 15°c is 0.01poise.

3. **a.** The rate of discharge Q of a Centrifugal pump is dependent upon the density ρ of the fluid, pump speed N in RPM, diameter of the impeller D, pressure P and viscosity of the fluid μ. Using Buckingham's $\pi - theorem.$ Show that $Q = ND^3\phi\left[\frac{P}{\rho N^2 D^2}, \frac{\mu}{\rho N D^2}\right]$. **08**

b. Starting from fundamentals, show that the velocity of propagation of elastic wave in an adiabatic medium is given by $C = \sqrt{\gamma RT}$ where C= velocity of the sound wave, γ= Ratio of specific heats, R= gas constant, T= Absolute Temperature. **06**